INCH-POUND

MIL-M-28787/253
AMENDMENT 2
28 April 1993
SUPERSEDING
AMENDMENT 1
18 April 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE BMA

This amendment forms a part of Military Specification MIL-M-28787/253(EC), dated 17 December 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, and 7 of this amendment.

PAGE 7

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence."

*4.5.1, line 5: Delete "B through J" and substitute "D through J".

4.5.1, line 8: Delete "100°C" and substitute "85°C".

*4.5.3, line 3: Delete "B through BB" and substitute "D through BB".

*4.5.3, line 6: Delete "B through CC" and substitute "D through CC".

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6.4, Output drive current (V_{DD} = 5 V), I_{OL} (V_{OL} = 0.4 V) at: Delete "100°C" and "0.36" and substitute "85° C" and "0.41" respectively.

6.4, Output drive current (V_{DD} = 5 V), I_{OH} (V_{OH} = 4.6 V) at: Delete "100°C" and "0.36" and substitute "85° C" and "0.41" respectively.

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FIGURE 6: Delete and substitute new figure 6 as printed on page 8 of this amendment.

PAGE 20

FIGURE 7: Delete and substitute new figure 7 as printed on page 9 of this amendment.

PAGE 23

FIGURE 8: Delete and substitute new figure 8 as printed on page 10 of this amendment.

PAGES 24 AND 25

TABLE IX: Delete and substitute new table IX as printed on pages 11 and 12 of this amendment.

PAGE 26

FIGURE 9: Delete and substitute new figure 9 as printed on page 13 of this amendment.

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FIGURE 10: Delete and substitute new figure 10 as printed on page 14 of this amendment.

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*FIGURE 16, ITEM C01 box: Delete "C01" and substitute "C06".

*FIGURE 16, NOTE 3, first line: Delete "(C01)" and substitute "(C06)".

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Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0205)

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TABLE I. Module electrical requirements.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage (negative)	IV	S	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage (positive)	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Output voltage (high level), IOH = 0	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E12	Output voltage (low level), IOL = 0	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F24	Output voltage (high level), VDD = 5 V	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G12	Output voltage (low level), VDD = 5 V	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H12	Output voltage (low level), VDD = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Output voltage (high level), VDD = 15 V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V

- See footnotes at end of table

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K24	Input current (high level)	XII	X	0	200	0	200	0	200	0	200	0	200	nA
L1-L24	Input current (low level)	XIII	X	0	-200	0	-200	0	-200	0	-200	0	-200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	100	µA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	100	µA
N1-N12	Transition time (tTLH), CL = 50 pF, VDD = 5 V	XV	S	20	200	20	200	22	220	15	150	23	235	ns
N13-N24	Transition time (tTLH), CL = 150 pF, VDD = 5 V	XV	Q	68	680	68	680	75	750	51	515	80	805	ns
P1-P12	Transition time (tTHL), CL = 50 pF, VDD = 5 V	XVI	S	20	200	20	200	22	220	15	150	23	235	ns
P13-P24	Transition time (tTHL), CL = 150 pF, VDD = 5 V	XVI	Q	68	680	68	680	75	750	51	515	80	805	ns
Q1-Q24	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XVII	S	25	250	25	250	27	275	19	190	29	295	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q25-Q48	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XVII	Q	45	450	45	450	49	495	34	340	53	530	ns
R1-R24	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XVIII	S	25	250	25	250	27	275	19	190	29	295	ns
R25-R48	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XVIII	Q	45	450	45	450	50	500	34	340	53	530	ns
S1-S12	Transition time (tTLH), CL = 50 pF, VDD = 10 V	XIX	Q	10	100	10	100	11	110	7	75	12	120	ns
T1-T12	Transition time (tTHL), CL = 50 pF, VDD = 10 V	XX	Q	10	100	10	100	11	110	7	75	12	120	ns
U1-U24	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXI	Q	10	100	10	100	11	110	7	75	12	120	ns
V1-V24	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXII	Q	10	100	10	100	11	110	7	75	12	120	ns

- See footnotes at end of table.

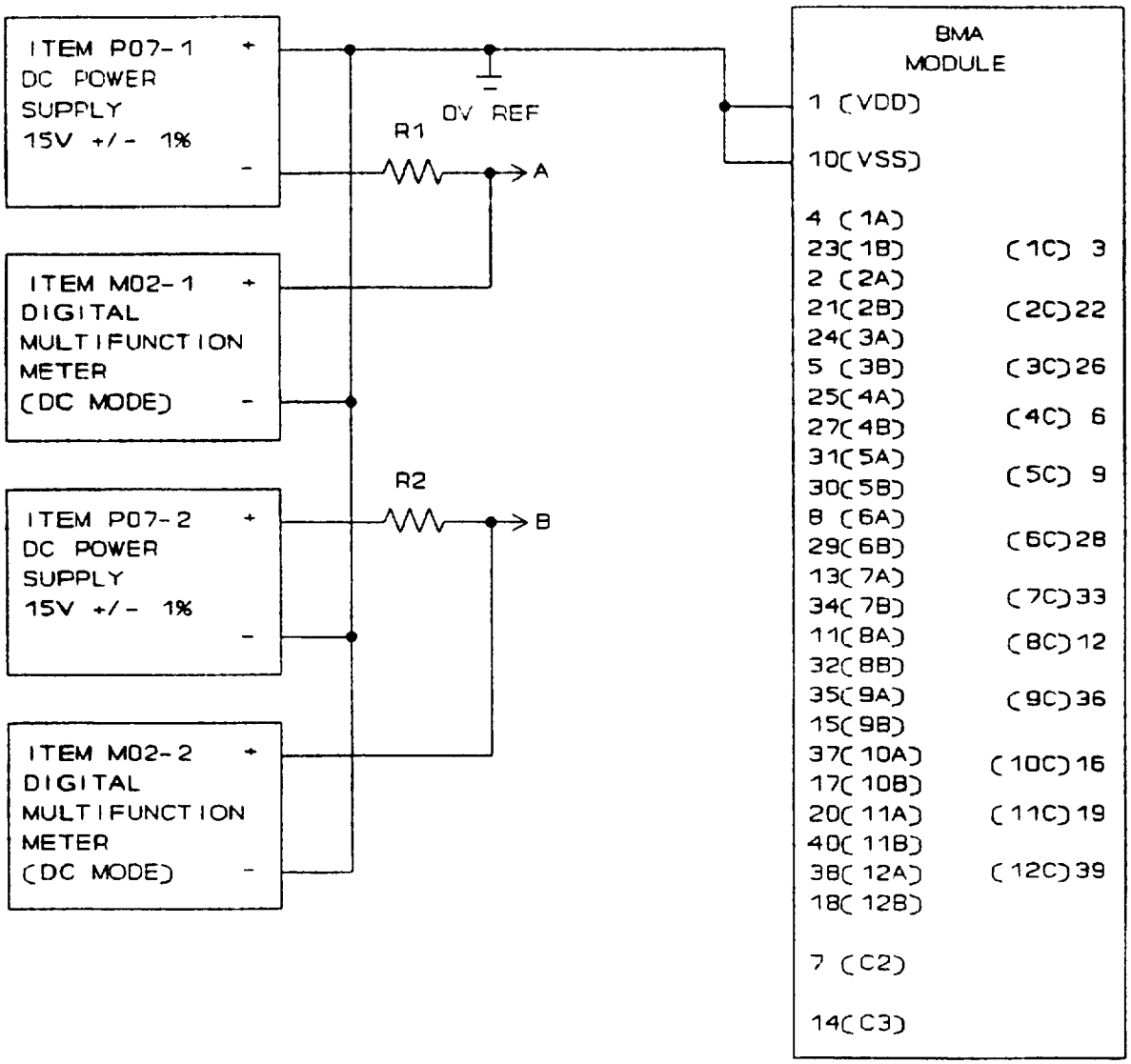
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TABLE I. Module electrical requirements - Continued.

Test Cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W1-W12	Transition time (tTLH), CL = 50 pF, VDD = 15 V	XXIII	Q	8	80	8	80	9	90	6	60	9	97	ns
X1-X12	Transition time (tTHL), CL = 50 pF, VDD = 15 V	XXIV	Q	8	80	8	80	9	90	6	60	9	97	ns
Y1-Y24	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXV	Q	8	80	8	80	9	90	6	60	9	97	ns
Z1-Z24	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXVI	Q	8	80	8	80	9	90	6	60	9	97	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	μF
BB1, BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

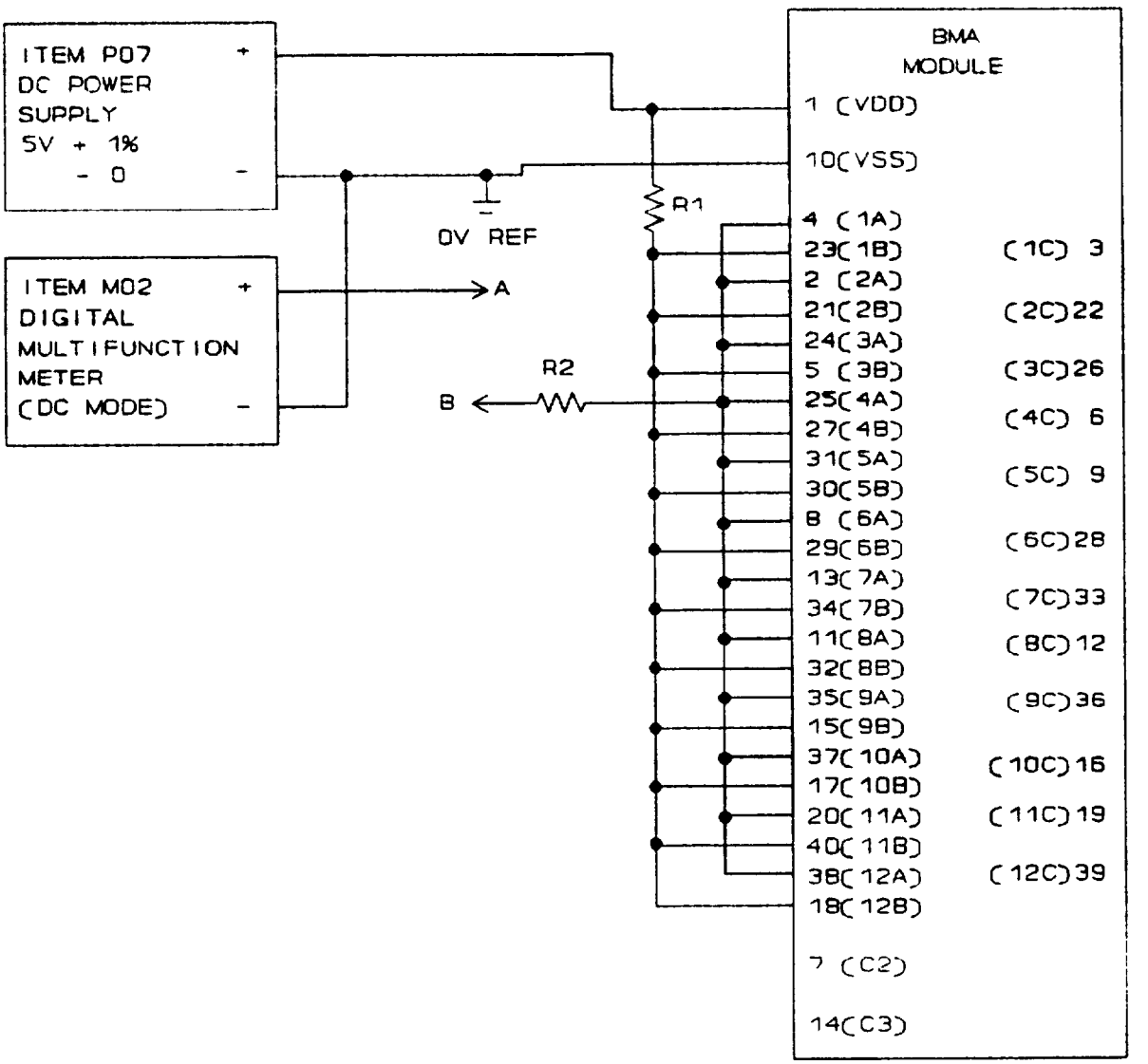
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NOTES:
1. R1 = 9.09 kΩ ± 1%, 1/8 W.
2. R2 = 13.7 kΩ ± 1%, 1/8 W.
3. Since items P07-1 and P07-2 and items M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

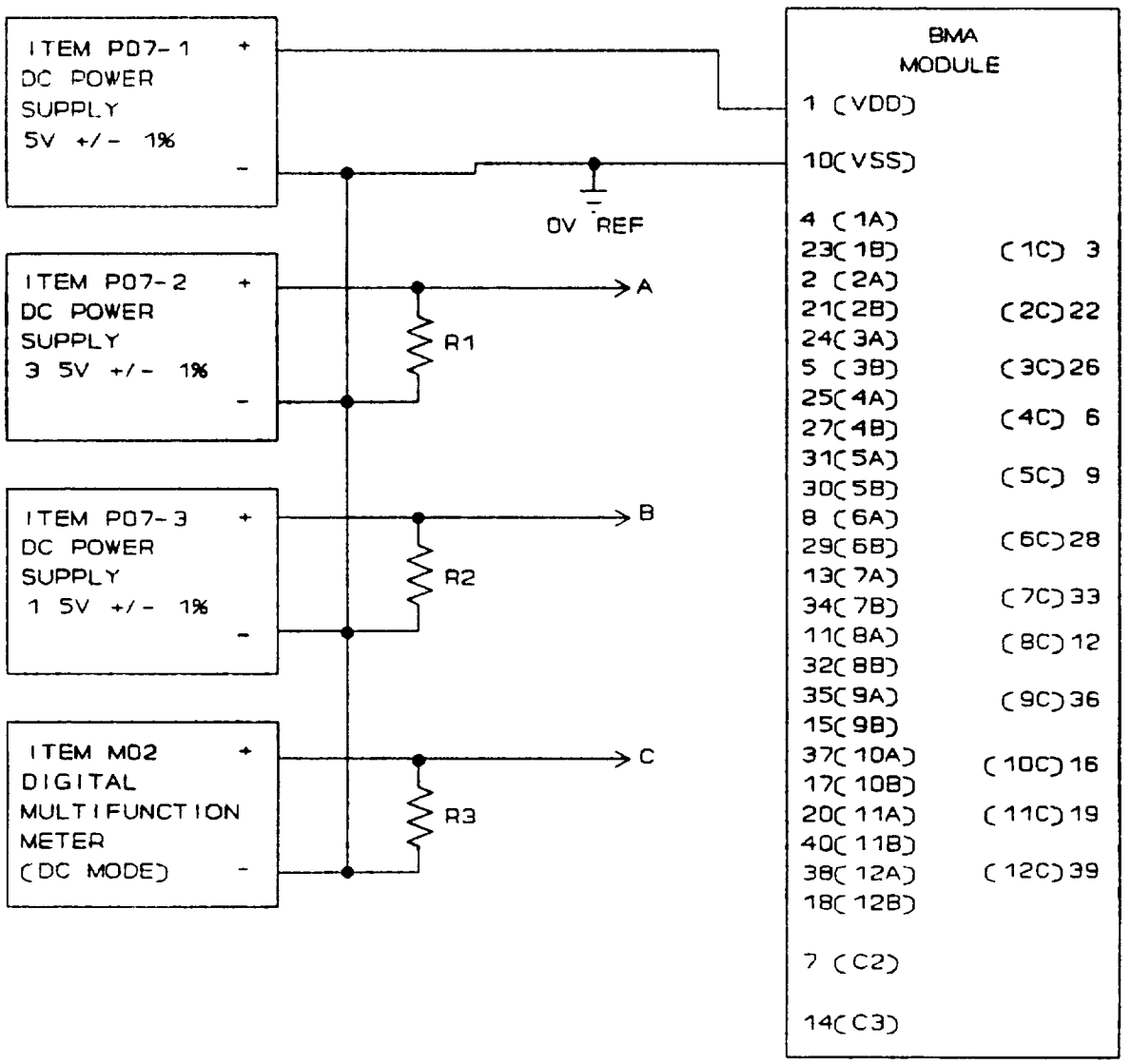
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NOTE: R1 = R2 = 1.5 kΩ ± 1%, 1/4 W.

FIGURE 7. Output voltage (high-level, IOH = 0 and low-level, IOL = 0) test circuit for tables VI and VII.

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NOTES:

1. R1 = R2 = 51.1 Ω ± 5%, 1/2 W.
2. R3 = At 0° and 25°C - 23.2 kΩ ± 1%, 1/8 W.
   = At 60°C - 27.4 kΩ ± 1%, 1/8 W.
   = At -55°C - 18.2 kΩ ± 1%, 1/8 W.
   = At 85°C - 30.1 kΩ ± 1%, 1/8 W.

FIGURE 8. Output voltage (high-level, VDD = 5 V) test circuit for table VIII.

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TABLE IX. Output voltage (low-level, VDD = 5 V) test.

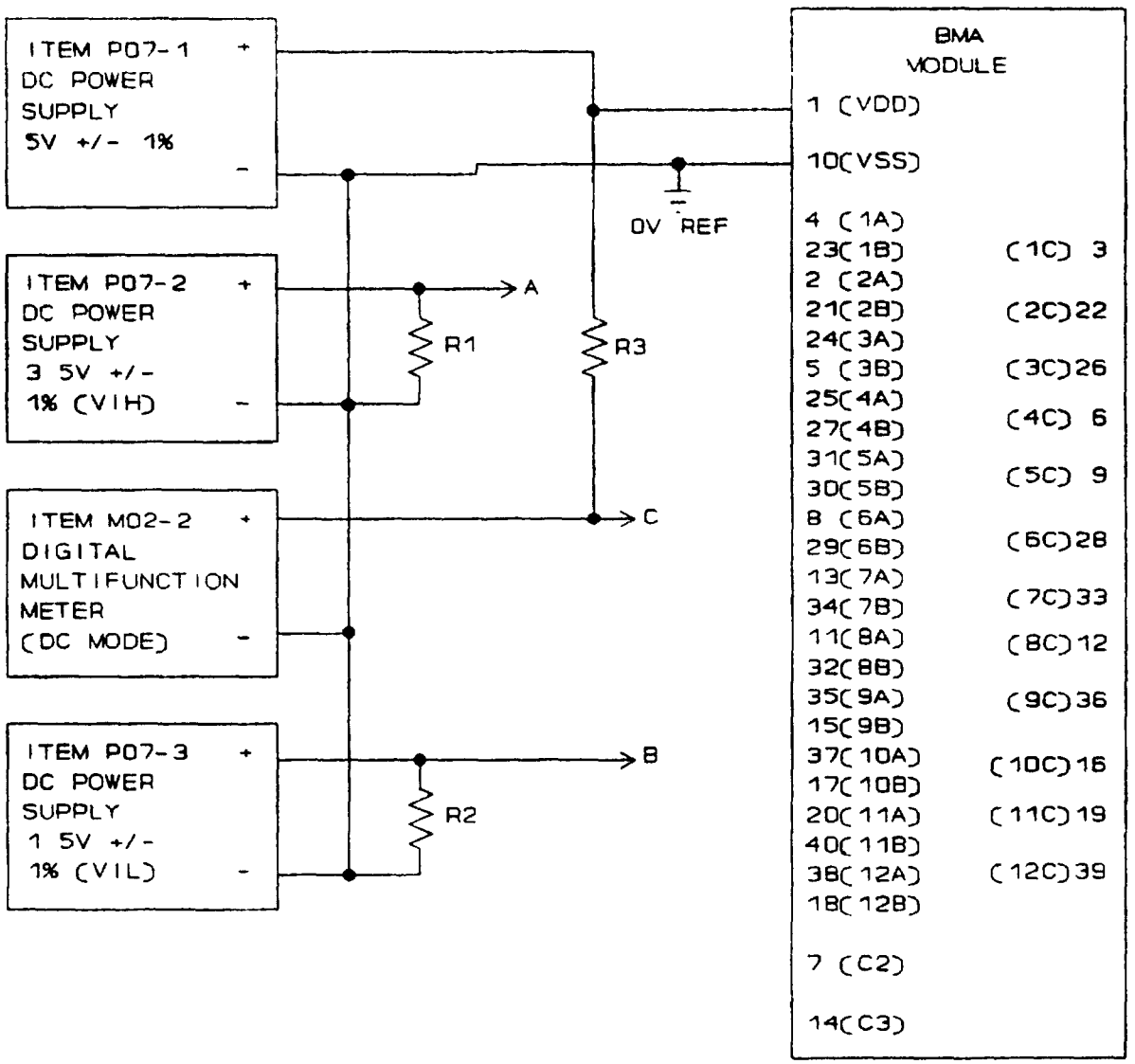
Test Cond	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1C Output voltage (low level), VDD = 5 V	4,23	2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	3	0.0	0.4	V	9
G2	2C	2,21	4,23,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	22				
G3	3C	24,5	4,23,2,21,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	26				
G4	4C	25,27	4,23,2,21,24,5,31,30,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	6				
G5	5C	31,30	4,23,2,21,24,5,25,27,8,29,13, 34,11,32,35,15,37,17,20,40,38,18	9				
G6	6C	8,29	4,23,2,21,24,5,25,27,31,30,13, 34,11,32,35,15,37,17,20,40,38,18	28				
G7	7C	13,34	4,23,2,21,24,5,25,27,31,30,8,29, 11,32,35,15,37,17,20,40,38,18	33				
G8	8C	11,32	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,35,15,37,17,20,40,38,18	12				
G9	9C Output voltage (low level), VDD = 5 V	35,15	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,37,17,20,40,38,18	36	0.0	0.4	V	9

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TABLE IX. Output voltage (low-level, VDD = 5 V) test - Continued.

Test Cond	Parameter	Connect			Initial limits 25°C ± 5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G10	10C Output voltage (low level), VDD = 5 V	37,17	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,20,40,38,18	16	0.0	0.4	V	9
G11	11C	20,40	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,38,18	19	—	—	—	—
G12	12C Output voltage (low level), VDD = 5 V	38,18	4,23,2,21,24,5,25,27,31,30,8,29,13, 34,11,32,35,15,37,17,20,40	39	0.0	0.4	V	9

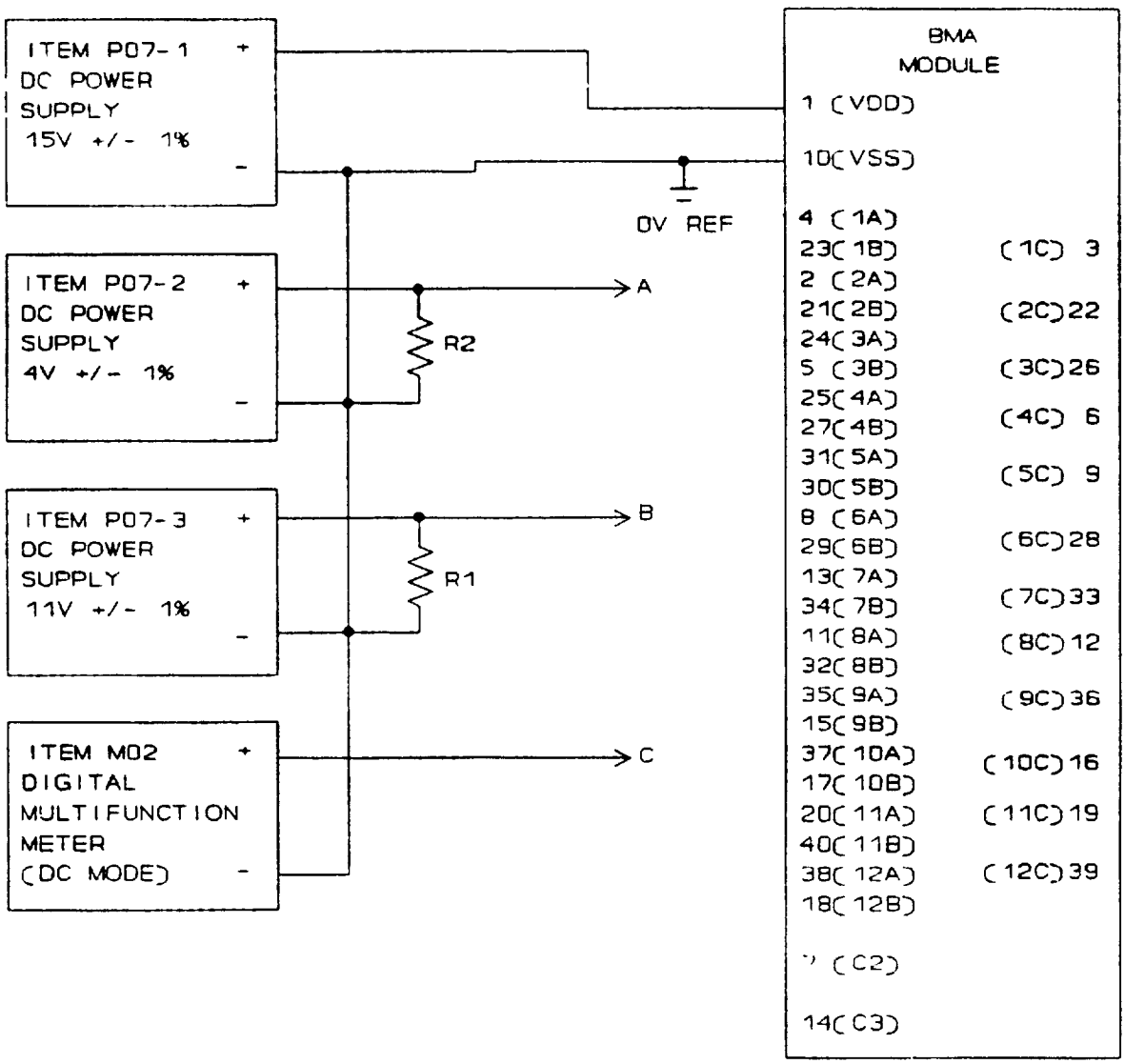
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NOTES:
1. R1 = R2 = 51.1 Ω ± 5%, 1/2 W.
2. R3 = At 0° and 25°C - 9.09 kΩ ± 1%, 1/8 W.
   = At 60°C - 10.5 kΩ ± 1%, 1/8 W.
   = At -55°C - 7.15 kΩ ± 1%, 1/8 W.
   = At 85°C - 11.8 kΩ ± 1%, 1/8 W.

FIGURE 9. Output voltage (low-level, VDD = 5 V) test circuit for table IX.

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NOTES:
1. R1 = 51.1 Ω ± 5%, 1/2 W.
2. R2 = 576 Ω ± 5%, 1/4 W.

FIGURE 10. Output voltage (low-level and high-level, VDD = 15 V) test circuit for tables X and XI.

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